

**Decent Coaching Classes**  
**VIII CBSE**  
**Mind Your Math - VI**

**Subject : Mathematics (Ch. 6 — We Distribute, Yet Things Multiply)**

**Marks : 30**

*Note : All questions are compulsory. Section C has internal choice in each question.*

**Section A — Multiple Choice Questions (1 mark each)**

1.  $(x + y)(x + y)$  is equal to : [1]  
(a)  $x^2 + y^2$  (b)  $x^2 + 2xy + y^2$  (c)  $x^2 - 2xy + y^2$  (d)  $2xy$
2. Using the distributive property,  $76 \times 101 = 76 \times 100 + 76 \times \underline{\hspace{2cm}}$ . [1]  
(a) 100 (b) 1 (c) 76 (d) 10
3.  $(x^2 + 1/x^2)(x^2 - 1/x^2)$  is equal to : [1]  
(a)  $x^4 - 1/x^4$  (b)  $x^4 + 1/x^4$  (c)  $x^2 - 1/x^2$  (d) None of these
4. Which identity is most suitable to evaluate  $96 \times 104$  quickly? [1]  
(a)  $(a + b)^2$  (b)  $(a - b)^2$  (c)  $(a + b)(a - b)$  (d) None of these
5.  $9x^2 + 6x + 1$  can be written as a perfect square : [1]  
(a)  $(3x + 1)^2$  (b)  $(3x - 1)^2$  (c)  $(9x + 1)^2$  (d)  $(x + 1)^2$
6. The value of  $47^2$  is : [1]  
(a) 2189 (b) 2209 (c) 2199 (d) 2219

**Section B — Very Short Answer Questions (2 marks each)**

7. Expand using an identity :  $(7y + 6)^2$  [2]
8. Find the product :  $(x^2 + 4)(x^2 + 4)$  [2]
9. Using a suitable identity, evaluate :  $96 \times 104$  [2]
10. Multiply using the distributive property :  $4x(5x^2 + 3x)$  [2]

**Section C — Short Answer Questions (4 marks each, with internal choice)**

11. Find the continued product :  $(x + 2)(x - 2)(x^2 - 4)$  [4]

**OR**

Find the continued product :  $(2x - 1)(2x + 1)(4x^2 + 1)$

12. Using a suitable identity, evaluate :  $(998)^2$  [4]

**OR**

Using a suitable identity, evaluate :  $(1005)^2$

13. Simplify :  $(3x + 2)^2 - (3x - 2)^2$  [4]

**OR**

Simplify :  $(2x/7 - y/2)^2 - (2x/7 + y/2)^2$

**Section D — Case Study Based Question (4 marks)**

14. A teacher wrote the following puzzle on the board for her students :

$$12x = 34^2 - 26^2$$

Based on the puzzle, answer the following questions :

- (a) Simplify  $34^2 - 26^2$  using a suitable identity. [2]
- (b) Hence find the value of  $x$ . [1]

(c) Verify your value of  $x$  by direct substitution in the equation. **[1]**